

AYE  
TECH HUB

• SOLAR ENERGY • LESSON 06

# Inverters

Explained

Lesson 06 of 30 | Solar Energy Program

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- What an inverter does — DC to AC conversion and MPPT tracking
- String vs microinverter vs hybrid vs off-grid inverter compared
- Key inverter specifications: efficiency, MPPT range, IP rating
- How to size an inverter for your solar panel array
- Anti-islanding, grid connection requirements and inverter settings

# The Solar Inverter — Heart of the AC System

A solar inverter does three critical jobs. First, it converts **DC electricity from the panels to AC electricity** that appliances and the grid can use. Second, it continuously runs **Maximum Power Point Tracking (MPPT)** — adjusting the panel operating point thousands of times per second to extract maximum power under changing irradiance, temperature, and shading conditions. Third, it provides **monitoring, protection, and grid management** functions that keep the system safe and compliant.

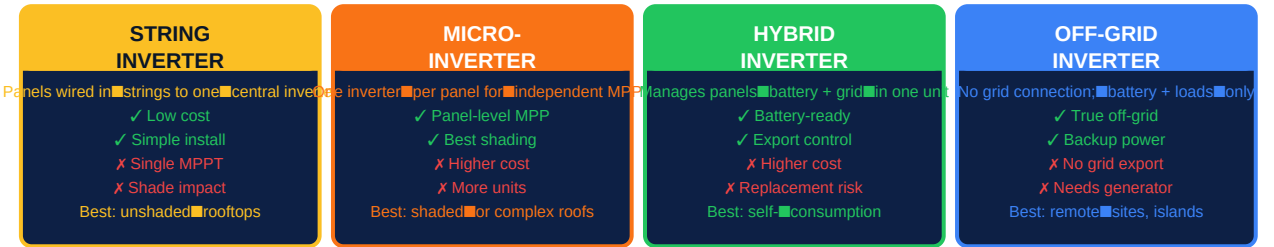


Figure 1 — Solar Inverter Types

| Inverter Type                     | MPPT Channels               | Typical Cost     | System Size    | Key Decision Factor                                         |
|-----------------------------------|-----------------------------|------------------|----------------|-------------------------------------------------------------|
| String inverter (standard)        | 1–3 MPPT inputs             | Lowest           | 1 kW – 100+ kW | Lowest cost; best for unshaded, simple rooftops             |
| String inverter (with optimisers) | 1–3 MPPT + panel optimisers | Medium           | 3 kW – 30 kW   | Shaded rooftops; want monitoring but not microinverter cost |
| Microinverter                     | 1 MPPT per panel            | Highest (per kW) | 0.3 kW – 15 kW | Maximum shade tolerance; complex roof; monitoring per panel |
| Hybrid (multi-mode)               | 2–4 MPPT + battery port     | High             | 3 kW – 30 kW   | Battery storage now or in future; backup power needed       |
| Off-grid inverter/charger         | 1–2 MPPT                    | Medium–High      | 0.5 kW – 30 kW | No grid connection; island systems; generator integration   |

**MPPT explained:** On a typical day, irradiance and temperature change continuously. The panel I-V curve shifts with every cloud. Without MPPT, the inverter would operate at a fixed voltage, missing 10–30% of available energy. MPPT continuously searches for the peak of the P-V curve (Maximum Power Point) and locks the operating voltage there — capturing every available watt.

# Inside the Inverter — How DC Becomes AC

Understanding the internal stages of an inverter helps you interpret datasheets, diagnose faults, and specify correctly. Every grid-tied inverter contains these five functional blocks in sequence.

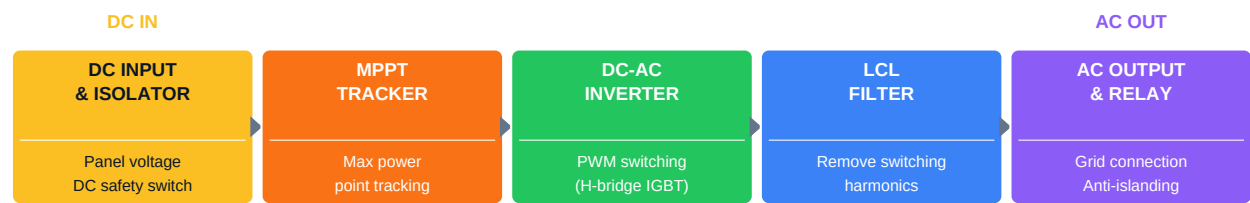


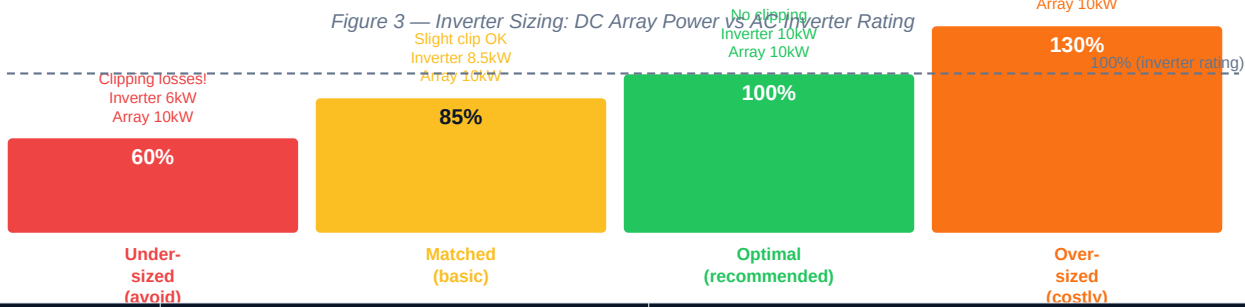
Figure 2 — Inverter Internal Block Diagram: DC Input to AC Output

| Key Inverter Specification                 | What It Means                                                                                  | Good Value                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Max DC input voltage (V_dc_max)            | Maximum open-circuit string voltage the inverter can safely accept. Must not be exceeded.      | 1 000 V (standard) or 1 500 V (utility scale). Design string so $V_{oc\_max} < V_{dc\_max}$ |
| MPPT voltage range (V_mpp_min – V_mpp_max) | DC voltage range within which MPPT operates. Panels must operate in this range.                | Wide range (e.g. 200–950 V) gives more flexibility in string design                         |
| Max DC input current per MPPT (A)          | Maximum string current the MPPT channel can accept. Must exceed string $I_{sc}$ .              | Size so string $I_{sc} \times 1.25 \leq \text{max input current}$                           |
| Rated AC output power (W)                  | Continuous AC power the inverter delivers to the grid/loads at nominal conditions.             | Size $\geq$ design load (off-grid) or $\geq 80\%$ of array peak DC (on-grid)                |
| European / CEC weighted efficiency (%)     | Real-world efficiency averaged across typical operating conditions — more realistic than peak. | 96–99% for quality string inverters; microinverters 95–97%                                  |
| Power factor / THD                         | Power factor: AC output quality. THD: harmonic distortion injected into grid.                  | PF $\geq 0.99$ ; THD $< 3\%$ (required by most grid codes)                                  |
| IP rating                                  | Ingress Protection against dust and water.                                                     | IP65 minimum for outdoor wall mount; IP54 for indoor/sheltered                              |

**Anti-islanding protection (mandatory for all grid-tied inverters):** If the grid fails, the inverter must detect this within 2 seconds and disconnect — preventing it from energising a dead grid where utility workers may be working. All grid-compliant inverters have this built in. It is tested during commissioning and must comply with local standards (IEEE 1547, VDE 0126, G98/G99).

# Inverter Sizing, Selection and Settings

Inverter sizing is the relationship between the DC power of your panel array and the AC rating of the inverter. **The DC array can be larger than the inverter's AC rating** — this is called DC oversizing or clipping — because panels rarely reach their full STC rating in real conditions. Most grid codes allow DC/AC ratios of 1.0–1.4.



| Sizing Step                 | Calculation                                                                         | Example (10 kWp system, 48V off-grid)                                                     |
|-----------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1. Determine DC array power | $P_{array} = N_{panels} \times P_{max \text{ per panel}}$                           | $25 \text{ panels} \times 400 \text{ Wp} = 10\,000 \text{ Wp (10 kWp)}$                   |
| 2. Calculate DC/AC ratio    | $DC/AC \text{ ratio} = P_{array} / P_{inverter\_AC}$                                | Target 1.0–1.2 → $10 \text{ kWp} / 1.1 = 9.1 \text{ kW AC inverter minimum}$              |
| 3. Check MPPT voltage range | $V_{mpp\_string}$ must fall within inverter MPPT range at all temperatures          | $8 \text{ panels} \times 40V = 320V \text{ Vmp}$ . Verify $320V \in [200\text{--}950V]$ ✓ |
| 4. Check max string voltage | $N_{panels} \times V_{oc\_STC} \times \text{temp correction factor} < V_{dc\_max}$  | $8 \times 41.5V \times 1.12 = 372V < 1000V$ ✓ (safe)                                      |
| 5. Check input current      | $\text{String } I_{sc} \times 1.25 \leq \text{inverter max input current per MPPT}$ | $10.5A \times 1.25 = 13.1A \leq 15A$ ✓                                                    |
| 6. Select brand and model   | Compare EFF, MPPT channels, warranty, local service                                 | SMA, Fronius, SolarEdge, Huawei, GoodWe — all suitable                                    |

## Grid Connection Settings and Commissioning

| Setting                                      | Typical Value                                       | Purpose                                                              |
|----------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------|
| Grid voltage range                           | ±10% of nominal (207–253V for 230V grid)            | Inverter disconnects if grid voltage moves outside this range        |
| Grid frequency range                         | 47.5–51.5 Hz (Europe); 59–61 Hz (USA)               | Disconnects on frequency excursion — protects equipment              |
| Export limit (feed-in limit)                 | Set to utility requirement (0W, 50%, or unlimited)  | Prevents exporting more than the utility allows                      |
| Time-of-use / self-consumption mode (hybrid) | Programme charge/discharge schedule for battery     | Maximise self-consumption; charge from cheap overnight tariff        |
| Anti-islanding standard                      | G98/G99 (UK), VDE 0126 (Germany), IEEE 1547 (USA)   | Confirms inverter complies with local grid code for safety           |
| Data logging / monitoring                    | Enable WiFi/RS485/LAN for cloud monitoring platform | Essential for performance tracking, fault diagnosis, warranty claims |

**What comes next — SOLAR-007: Charge Controllers — deep dive into MPPT and PWM charge controllers, charge stage management (bulk/absorption/float/equalize), how to match a charge controller to your battery**

*bank, and advanced MPPT controller settings.*